GAME DESIGN

UNIVERSIDADE da MADEIRA

COURSE OVERVIEW

UNIVERSIDADE da MADEIRA

- Introduction to class syllabus
- Course goals
- Assessment breakdown (e.g., participation, prototyping, game analysis, final project)
- Biography for the class
- Key deadlines and deliverables

PLAY

UNIVERSIDADE da MADEIRA

Playing games is considered na importante activity as it can developed problem-solving skills by analyzing solutions, increase competency and autonomy by allowing users to make decisions, and improve social skills by playing on collaborative environments.

Hewett,K.J et al, 2020 and Yilmaz E. 2023

Playing is a natural activity for humans, specially children. Play is na intrinsic motivation that can also help children's development intellectually, socially, and physically.

(Devrizal et al., 2019).

Hewett, K. J., Zeng, G., & Pletcher, B. C. (2020). The acquisition of 21st-century skills through video games: Minecraft design process models and their web of class roles. Simulation & Gaming, 51(3), 336-364.;Yılmaz, E., Griffiths, M.D. Children's social problem-solving skills in playing videogames and traditional games: A systematic review. Educ Inf Technol 28, 11679–11712 (2023). https://doi.org/10.1007/s10639-023-11663-2

GAME

UNIVERSIDADE da MADEIRA

Not standardize definition…

A game is a type of play activity conducted in the context of a pretended reality in which the participant(s) try to achieve at least one arbitrary, nontrivial goal by acting following rules.

Ernest Adam (2014)

An interactive experience where oneor several players need to reach a goal by taking some actions (following a set of rules)

Carla Fisher (2014)

Adams, E. (2014). Fundamentals of game design. Pearson Education; Fisher, C. (2014). Designing games for children: Developmental, usability, and design considerations for making games for kids. Routledge.

GAME

UNIVERSIDADE da MADEIRA

"A game is a problem-solving activity, approached with a playful attitude"

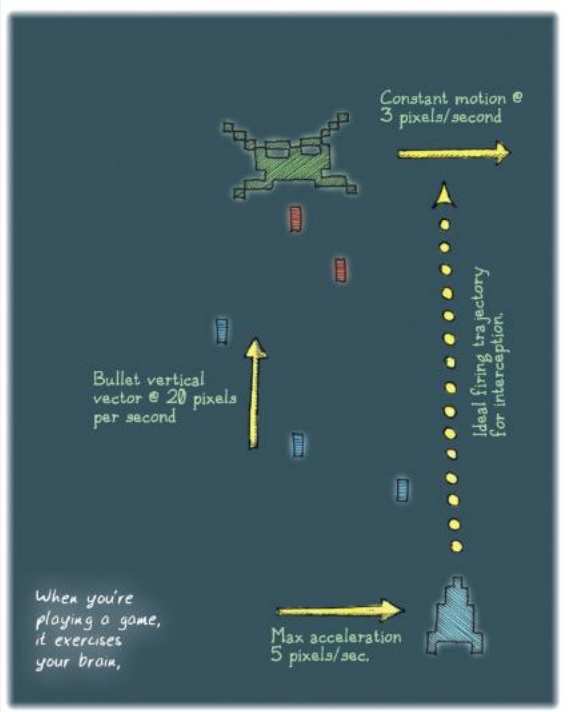
Jessy Schell (2008)

A game as a system in which players engage in an artificial conflict, defined by rules, that results in a quantifiable outcome

Katie Salen and Eric Zimmerman (2003)

J. Schell. The Art of Game Design: A book of lenses. CRC press, 2008 (cit. on pp. 24,26, 28, 32–34, 100, 102); K. S. Tekinbas and E. Zimmerman. Rules of play: Game design fundamentals. MIT press, 2003 (cit. on p. 24).

*"Fun from games arises out of mastery. It arises out of comprehension. It is the act of solving puzzles that makes games fun."***Theory of Fun for Game Design, 2nd Edition**, *by Raph Koster*



"When a game stops teaching us, we feel bored. Boredom is the brain casting about for new information".

Theory of Fun for Game Design, 2nd Edition, *by Raph Koster*

PLAYER

Player is the participant that actively plays or Interacts with the system.

Psychological theories that can explain a specific reaction, emotion, or affect from the interaction between a game and its player:

- *Theory of **flow** or the "theory of optimal experience"* *(Csikszentmihalyi, 1998)*
- ***Presence** is a psychological sense of "being there"* (M. Slater et al., 1996)
- *The theory of **fun** (Raph Koster, 2013)*

M. Slater, V. Linakis, M. Usoh, and R. Kooper. “Immersion, presence and performance in virtual environments: An experiment with tri-dimensional chess”. In: Proceedings of the ACM symposium on virtual reality software and technology. 1996, pp. 163–172 (cit. on p. 25).; L. A. Beck. Csikszentmihalyi, Mihaly.(1990). Flow: The psychology of optimal experience. 1992 (cit. on p. 25); R. Koster. Theory of fun fo game design. "O’Reilly Media, Inc.", 2013 (cit. on pp. 25, 26).

GAME ENGAGEMENT

UNIVERSIDADE da MADEIRA

Intrinsic motivation: Doing an activity because is interesting and satisfying

Extrinsic motivation: Doing an activity because of receiving rewards or avoid punishment

These factors can potentially anticipate and predict positive players outcomes and engagement with a game, from .55 sec

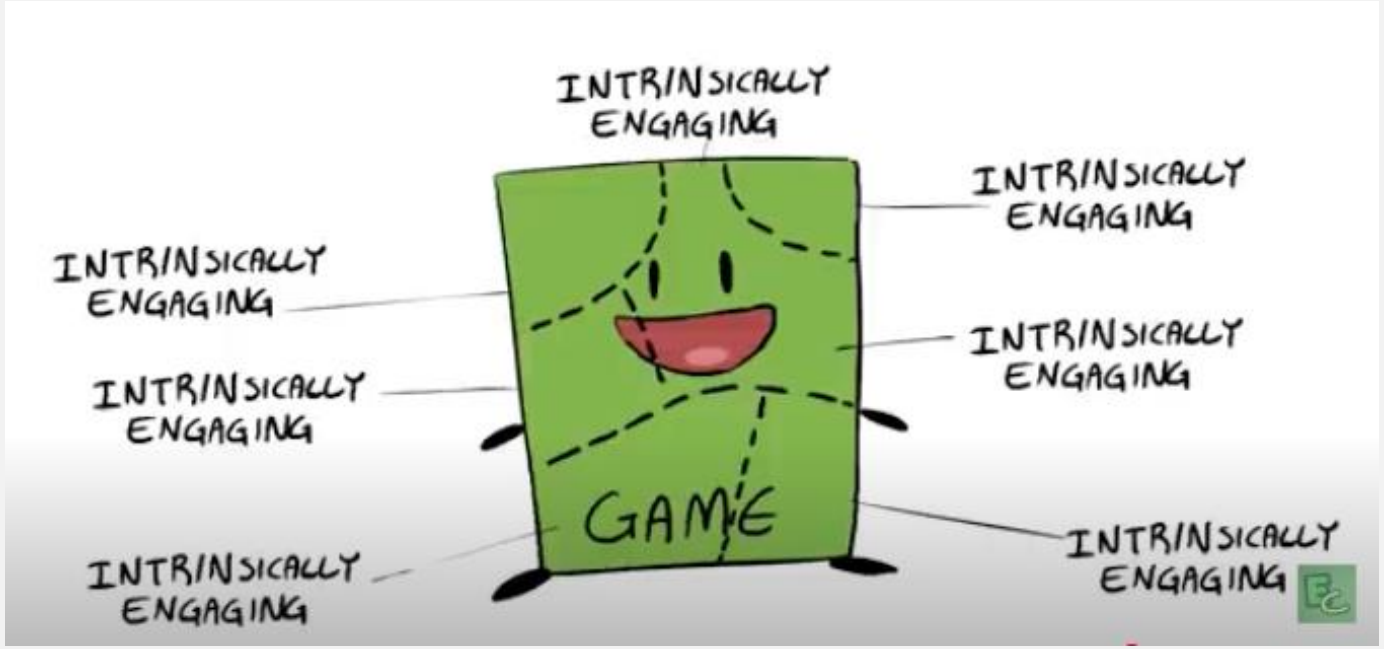
https://www.youtube.com/watch?v=h86g-XgUCA8

Deci and R. M. Ryan. Handbook of self-determination research. University, Rochester Press, 2004 (cit. on p. 35).

GAME ENGAGEMENT

UNIVERSIDADE da MADEIRA

Why am I playing?!



https://www.youtube.com/watch?v=h86g-XgUCA8 – from .55 sec

GAMIFICATION

UNIVERSIDADE da MADEIRA

It is considered the process of taking game elements or game like features (such as point, badges, levels, and missions) and applying them to nongame contexts, such as office tasks, classroom activities, etc. with the goal of, increasing productivity in the workplace, enhancing learning or improve a behaviour.

It can also increase player's autonomy, confidence and self-determination through intrinsic and extrinsic motivation

E. D. Mekler, F. Brühlmann, A. N. Tuch, and K. Opwis. "Towards understanding the effects of individual gamification elements on intrinsic motivation and performance". In: Computers in Human Behavior 71 (2017), pp. 525–534. url: https://doi.org/10.1016/j.chb.2015.08.048 (cit. on p. 36).

GAME DYNAMICS

UNIVERSIDADE da MADEIRA

Define by Hunicke as the behavior and actions that emerge from a game's rules and gameplay mechanics.

Game designers must anticipate patterns of behavior and actions to determine how players will engage with the game to motivate them to keep playing.

There are four critical components of game dynamics: ***sensation, fantasy, challenge and fellowship.***

R. Hunicke, M. LeBlanc, R. Zubek, et al. "MDA: A formal approach to game design and game research". In: Proceedings of the AAAI Workshop on Challenges in Game AI. Vol. 4. 1. San Jose, CA. 2004, p. 1722 (cit. on pp. 28, 30, 36).

TYPE OF GAMES

UNIVERSIDADE da MADEIRA

By Format / Medium:

Analog games: “real” or physical tangible games that not rely on digital technology.

The dymamics of the game involve physical interaction, social engagement with direct human interaction. E.g. Board (Catan, Monopoly), card (Uno, Poker), dice (Zombie Dice, Liar’s dice), tile –laying (Jenga),etc.

M. Sousa. “Modern Serious Board Games: modding games to teach and train civil engineering students”. In: 2020 IEEE Global Engineering Education Conference (EDUCON). IEEE. 2020, pp. 197–201 (cit. on p. 24).

TYPE OF GAMES

UNIVERSIDADE da MADEIRA

Analog games play part of our culture and social interactions

ANALOG GAMES

UNIVERSIDADE da MADEIRA

Analog, conventional or physical games offer the chance to develop cost-effective and easily adaptable experiences.

Can:

- Foster engagement
- Enhance player satisfaction
- Opportunities for enjoyable, social, flexible, and cost-effective experiences
- Facilitating knowledge acquisition
- Contribute to behavioral change

ANALOG GAMES

UNIVERSIDADE da MADEIRA

- Visible, tangible rules
- No timer: an independent referee sets the pace of the game
- The game world: Fantasy and imaginary worlds are constructed in the player's imagination.

Adams, E. (2014). *Fundamentals of game design*. Pearson Education.

ANALOG GAMES - ACTIVITY

UNIVERSIDADE da MADEIRA

Exploring Game Dimensions (work group – max 3 persons):

List and briefly describe ten analogs games, indicating the objective, risk/reward, and rules.

For each of the ten games listed above, classify them according to its dimension:

*physical (required body movement or dexterity)
*cognitive (requires thinking, planning or memory)
*emotional (evokes or relies on emotional reactions or interpersonal interactions)
*social (part of cultural or social interaction, involving cooperation, competition, or group dynamics)

Transform
– choose one game and change it to explore several of the dimensions above.
- explain the rules and justify your choice

DIGITAL GAMES

UNIVERSIDADE da MADEIRA

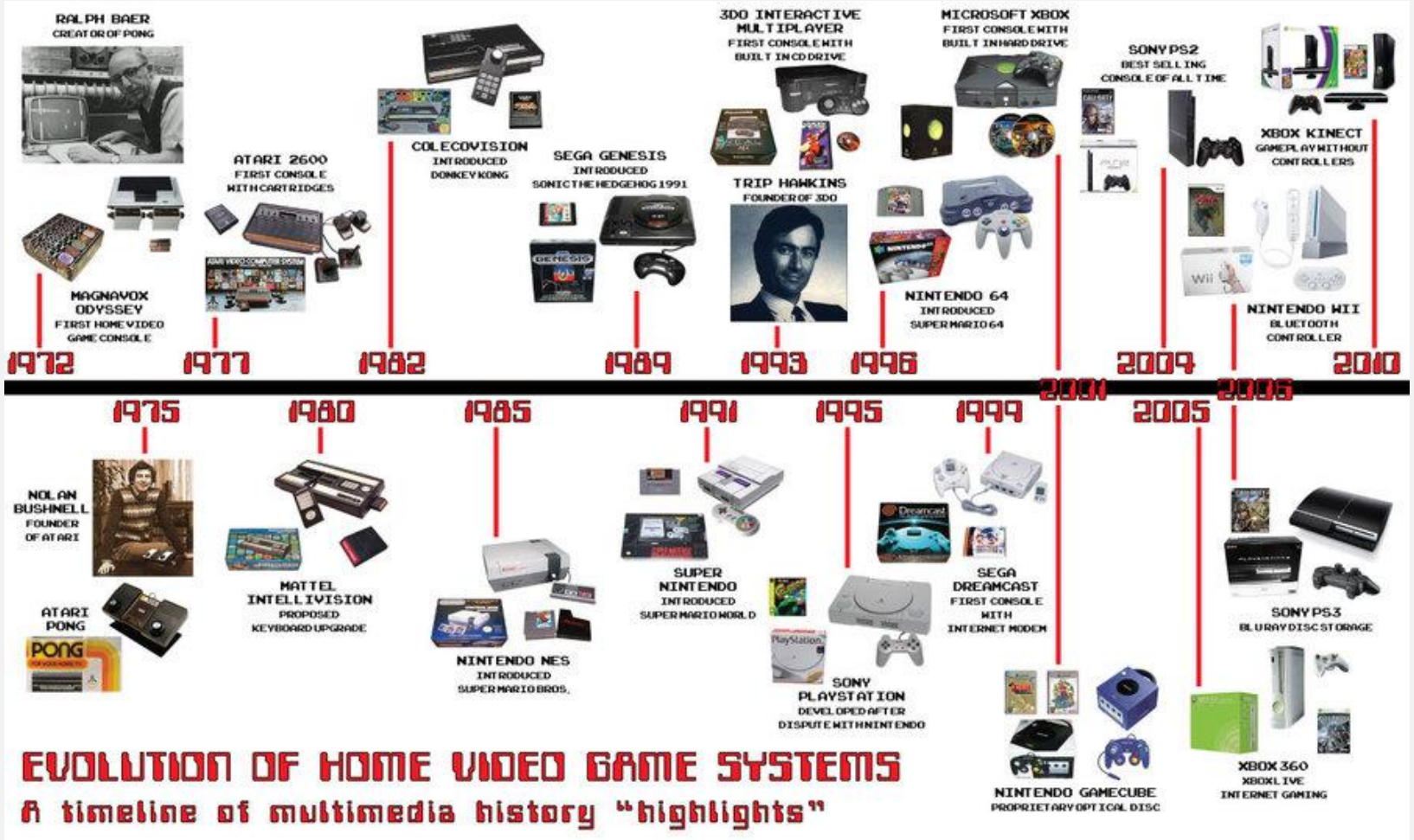
Digital games have been conceptualized as particular types of digital environments or as forms of virtual worlds.

According to Grant Tavinor, this type of game can be best defined according to two conditions: the **digital/visual** médium condition and the **entertainment condition**, both of which should provide a rule-bound gameplay and interactive fiction

G. Tavinor. "Definition of videogames". In: Contemporary Aesthetics (Journal Archive) 6.1 (2008), p. 16 (cit. on p. 25).

BRIEF HISTORY

UNIVERSIDADE da MADEIRA



Suominen, J. (2017). How to present the history of digital games: Enthusiast, emancipatory, genealogical, and pathological approaches. *Games and Culture*, *12*(6), 544-562.

BRIEF HISTORY

UNIVERSIDADE da MADEIRA

https://www.youtube.com/watch?v=h4CSGuiLik8

GAMES ACCORDING TO ITS PURPOSE

UNIVERSIDADE da MADEIRA

Entertaining Games: created with the only purpose of entertainment and fun

Serious Games: described as games with a purpose that are not intended to be played solely for amusement and are often designed to make beneficial but challenging activities more fun. Use in the educational, instructional realms.

Persuasive Games: raise awareness or change attitudes and behaviours (e.g. to create awareness about climate change).

ASSIGNMENT

UNIVERSIDADE da MADEIRA

Provide a list of as many conventional games you have interacted with. It can be from your childhood, family gathering, etc.

Select 1 and briefly explain the rules of the games.

Identify what factors made the game engaging

ASSIGNMENT

UNIVERSIDADE da MADEIRA

Write a report, max 550 words with the following:

List and briefly describe ten games from one of these decades: 70s, 80s, 90s, or 2000s indicating the objective, technology and rules.

Recreate one of the games.

Think on the platform, genres, commands, cultural theme or aesthetics, technological limitations for the decade.

Provide: game name, concept, mechanics, why is fun, what makes it a hit.

Reflect: Is there any game in the actual context that has the same game play mechanics?

Pitch the new game that would fit one of these decades (1 minute).

MIX REALITY GAMES

UNIVERSIDADE da MADEIRA

Virtual Reality (VR): create a fully immersive, three-dimensional environment in which players can move, look around, and interact with virtual objects as though they were physically present through special equipment (head-mounted display (HMD) and often uses motion-tracked controllers or body sensors).

Augmented Reality (AR): integrate virtual elements with the physical environment, allowing players to interact with both simultaneously through AR-enabled devices (like smartphones, tablets, or AR headsets).

MIX REALITY GAMES

UNIVERSIDADE da MADEIRA

Virtual Reality (VR) Games

Meta Quest, PC VR

PlayStation VR2, PC VR

Meta Quest, PS VR2, PC VR

AUGMENTED REALITY GAMES

UNIVERSIDADE da MADEIRA

UNIVERSIDADE da MADEIRA

GAME DESIGN

GAME DESIGN

UNIVERSIDADE da MADEIRA

Game design is a process of creating the content and rules of a game using a user centric approach. This process creates goals that a player feels motivated to reach and rules they must follow, making meaningful decisions to pursue those goals.

B. Brathwaite and I. Schreiber (2009)

"A game designer is a person who designs games. But more importantly, a game designer is someone who designs play."

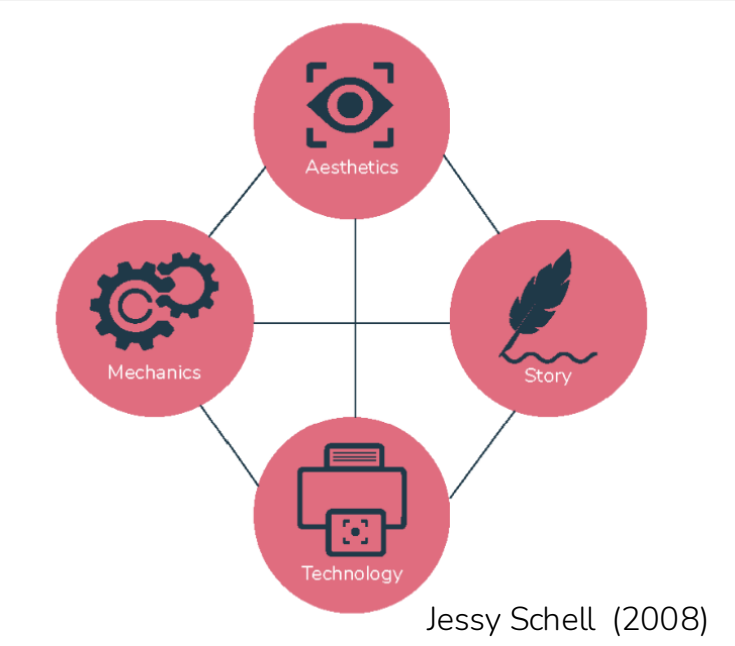
Jessy Schell (2008)

B. Brathwaite and I. Schreiber. Challenges for game designers. Course Technology/Cengage Learning Boston, Massachusetts, 2009 (cit. on p. 27), **The Art of Game Design: A Book of Lenses, 2008.**

GAME DESIGN

UNIVERSIDADE da MADEIRA

Elemental Tetrad - framework



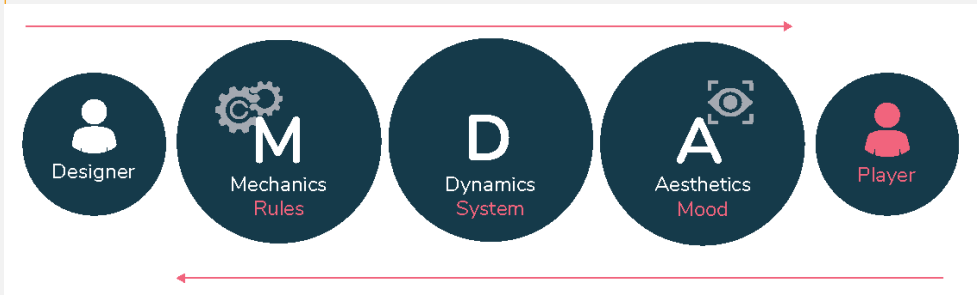
The Art of Game Design: A Book of Lenses, 2008.

These elements create a specific experiential flow by being interrelated and connected and having equal importance within the context of the game.

GAME DESIGN

UNIVERSIDADE da MADEIRA

Hunicke et al. - framework



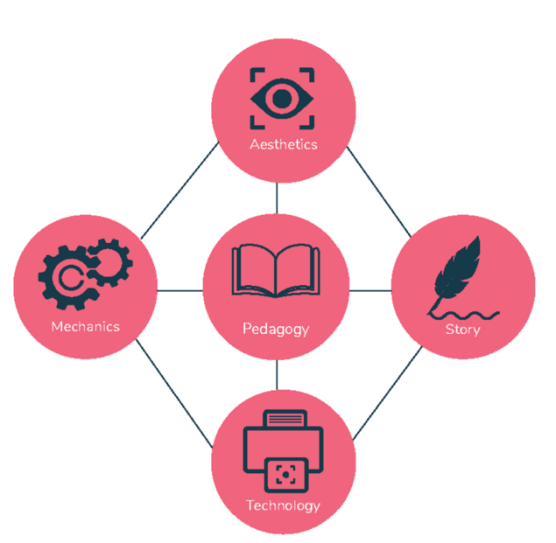
Dynamics include artifact behavior (usability issues like lag), player-player interaction (e.g., cooperation), and artifact-player interaction (e.g., explosion).

R. Hunicke, M. LeBlanc, R. Zubek, et al. "MDA: A formal approach to game design and game research". In: Proceedings of the AAAI Workshop on Challenges in Game AI. Vol. 4. 1. San Jose, CA. 2004, p. 1722 (cit. on pp. 28, 30, 36).

GAME DESIGN

UNIVERSIDADE da MADEIRA

George Kalmpurtzis. - framework



Likewise, this approach follows the four basic elements identified by Jessy Schell; however, when designing games for education, a fifth element needs to be part of this balance: the pedagogics (Kalmpourtzis, 2018b).

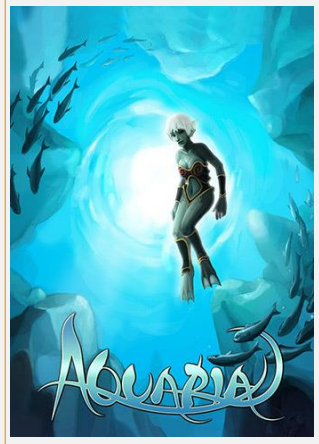
Kalmpourtzis, G. (2018). *Educational Game Design Fundamentals: A journey to creating intrinsically motivating learning experiences*. AK Peters/CRC Press.

AESTHETICS

UNIVERSIDADE da MADEIRA

The aesthetics of a game is defined as "the **desirable emotional responses evoked in the player** when he/she interacts with the game system"

The aesthetics are also considered as having the most direct relationship with the user since it provides the visuals and tone of the game (Schell, 2008).



R. Hunicke, M. LeBlanc, R. Zubek, et al. "MDA: A formal approach to game design and game research". In: Proceedings of the AAAI Workshop on Challenges in Game AI. Vol. 4. 1. San Jose, CA. 2004, p. 1722 (cit. on pp. 28, 30, 36).

AESTHETICS – ART STYLES

UNIVERSIDADE da MADEIRA

Pixel Art

Realism

Flat

vector

Cutout

Geometric

Monocromatic

ELEMENTS OF GAME ART STYLES

UNIVERSIDADE da MADEIRA

https://www.youtube.com/watch?v=D5BcoOCLur4

GAME CHARACTERS

UNIVERSIDADE da MADEIRA

Character design is fundamentally the process of developing a brand-new, unique persona for a medium, like a movie, comic book, or digital game [231]. The characters serve as a window allowing the viewer to witness a variety of places and events, so stories are perceived via them

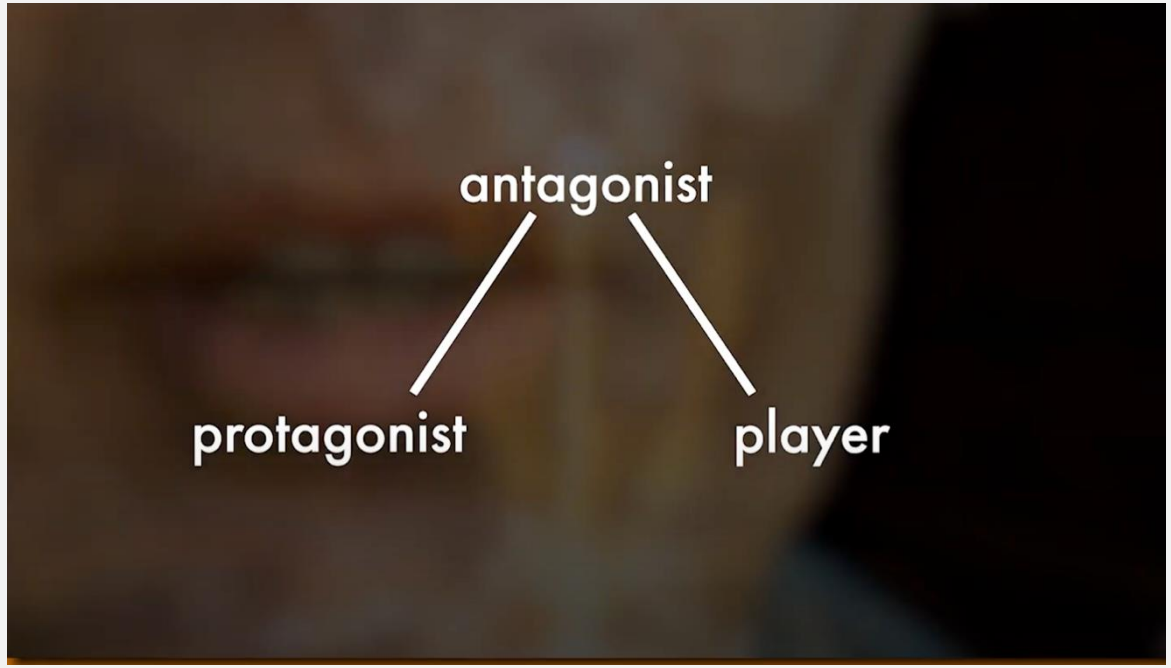
In order to elicit emotion and investment in the plot, characters must establish a relationship with the audience.

M. Nieminen. "Psychology in character design: Creation of a Character Design Tool". In: (2017) (cit. on p. 31).

GAME CHARACTERS - AGENCY

UNIVERSIDADE da MADEIRA

Character agency – create affects (empathy, fear, inspiration, humor, sadness, anger, etc) with your characters



https://www.youtube.com/watch?v=0tECLmhyDbs

CHARACTER DESIGN- IN CLASS INDIVIDUAL ACTIVITY

UNIVERSIDADE da MADEIRA

Create a main character (hero) who is is absurdly committed to one of this roles:

a) Explodes with Rage at Acts of Kindness- Furious when someone helps them carry something. Screams, "HOW DARE YOU LOVE ME."

b) Fear of Feeling Happy- Every time they start to feel joy, they self-sabotage in a dramatic opera of despair.

Define: name, backstory, core motivation, visual appearance

Create a nemesis and describe also name, backstory, visual appearance and core motivation

Translate their emotional flaw into the gameplay mechanics and explain the game concept.

Pitch: Justify your choices and explain your concept in 1 min.

ASSIGNMENT- GROUP WORK

UNIVERSIDADE da MADEIRA

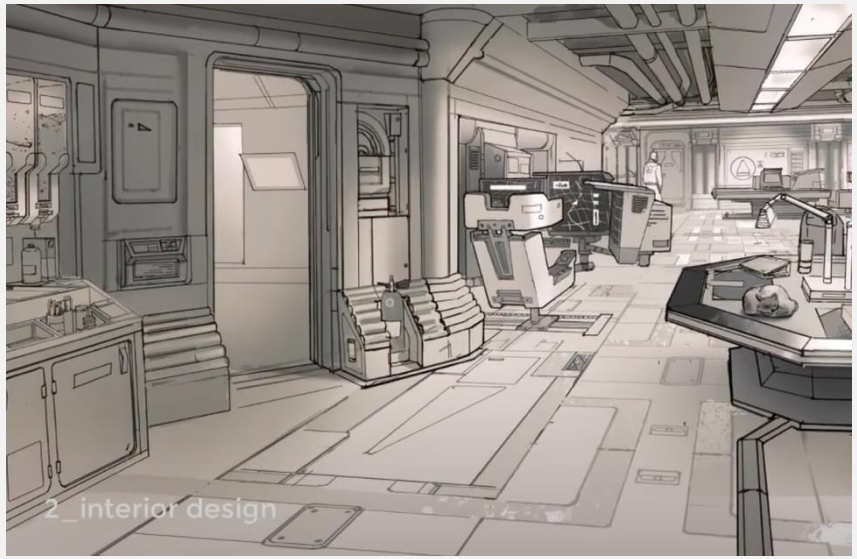
Create a board game that while entertaining can create awareness about one of the following topics: sustainability, mental health, or inclusion

Deliverables: name, rules of the game, concept art, and paper prototype

GAME SETTING/ENVIRONMENTS

UNIVERSIDADE da MADEIRA

The setting of a role-playing game is essentially a vast playground to adventure in.



https://www.youtube.com/watch?v=khlX1RMQbW0

STORYTELLING

UNIVERSIDADE da MADEIRA

Fundamental mode of human communication and remains a vital component of cognitive, emotional, and social development.

Storytelling involves a dynamic interaction between the narrative, the narrator, the medium, and the audience.

STORY: consists of all the events which are to be depicted.

Cobley, P. (2013). Narrative. Routledge.

GAMES AND NARRATIVES

UNIVERSIDADE da MADEIRA

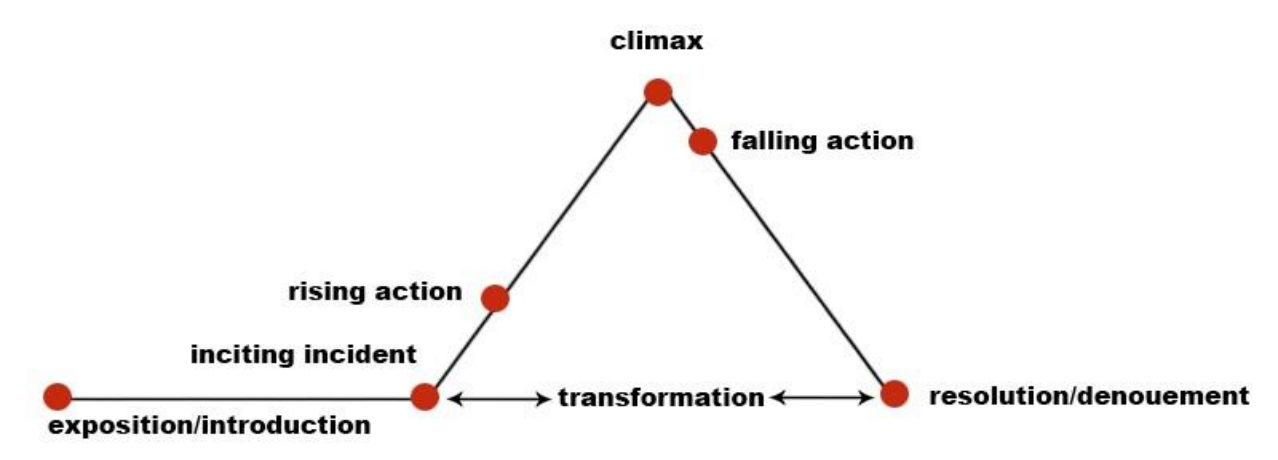
A game designer defines the rules that govern how a particular game should be played by setting goals and imposing interesting obstacles that prevent the player from immediately achieving those goals.

These two mechanisms goals and obstacles create the conflict that is at the heart of every story irrespective of the medium, be it film, animation, novels, or video games.

Only with video games it is the player, and not the protagonist of the story, who must succeed through a mix of creative strategy, luck, and endurance.

If there is an opportunity this class can be a join effort with Visual Narrative II – (Narrative and games)

STORY STRUCTURE



Freytag's pyramid

Joseph Campbell

What is the story I want to tell?

What kind of audience participation do I want or need?

How much is based in the real world vs a fictional world?

How will I deliver the story?

How will audience participation affect the story over time?

STORYTELLING CLASS ACTIVITY

UNIVERSIDADE da MADEIRA

Select a well know story from your childhood and change it, to the point of view of another character or even an object contain within the story.

Think of a game play and the mechanics that you could use to convey the story.

Pitch your idea in a 1-minute presentation

MECHANICS

UNIVERSIDADE da MADEIRA

Game mechanics is another concept without a generally accepted definition in the industry and academics; however, research agrees that there is an interrelationship between mechanics and player behavior.

”The means that guides the player into a particular behavior by constraining the space of possible plans to attain goals”

(Järvinen, 2007)

MECHANICS

UNIVERSIDADE da MADEIRA

A MECHANIC is a rule about how a game works. The A button makes Mario jump is a mechanic. So are the rules characters walk at one meter per second, pawns capture diagonally, and players alternate taking turns.

In board games, mechanics are written in the rulebook.

MECHANICS

UNIVERSIDADE da MADEIRA

Types of Game Mechanics

- Primary Core Mechanics
  - Action(s) that lead participant(s) directly to a desired state (often the end state)
  - Available throughout the game
  - Easy to enact by participants

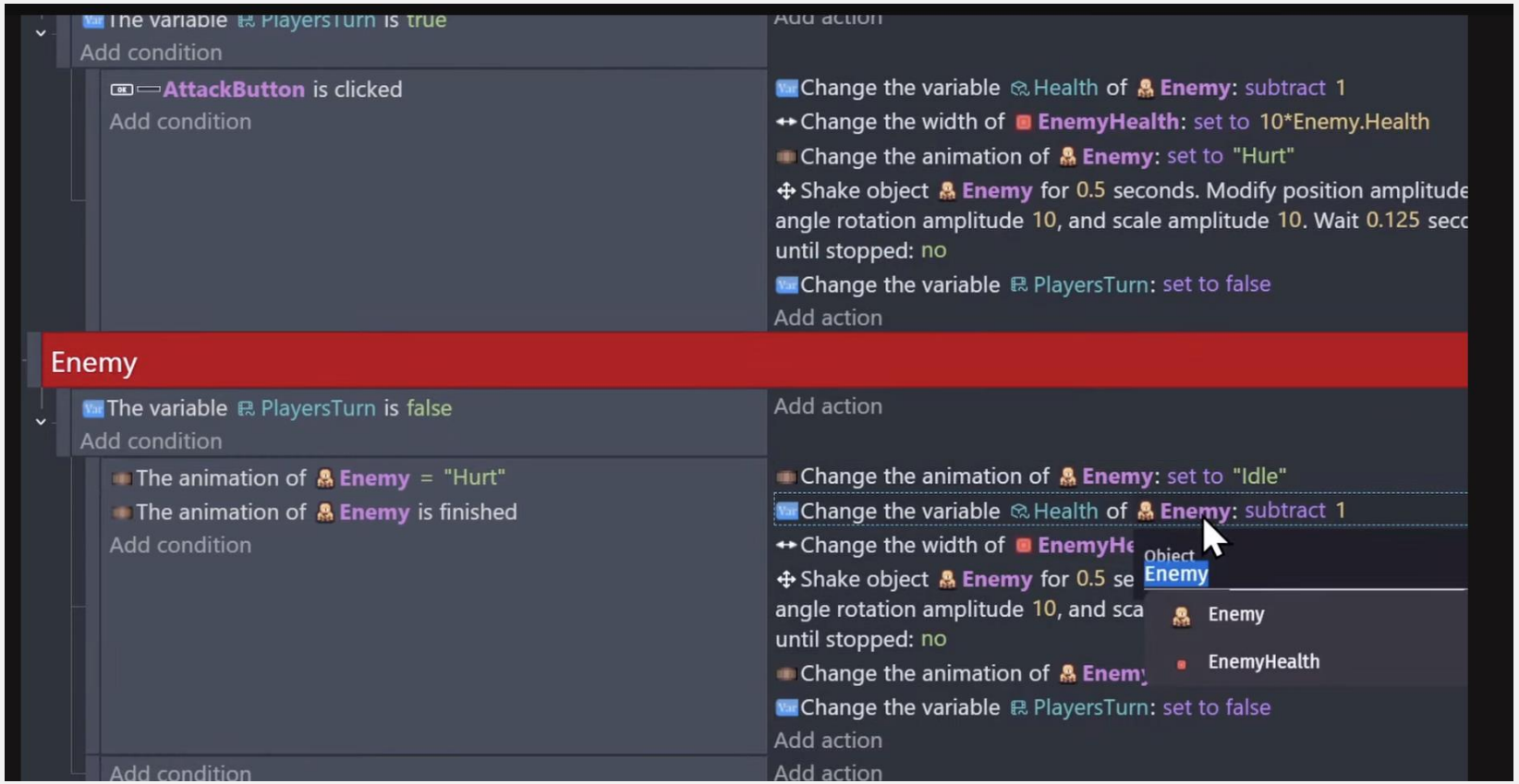
Mario Kart

PacMan

Katamari Damacy

https://www.youtube.com/watch?v=C-Ym5zlU9Uk

MECHANICS



https://www.youtube.com/watch?v=rk_NC05mLfM

GAMEPLAY

UNIVERSIDADE da MADEIRA

Challenges create tension and drama. At the simplest level, presenting players with a challenge amounts to asking the question, “Can you do it?” They’ll enjoy trying to prove that they can.

CHALLENGE TYPE	FAMILIAR EXAMPLE
Physical Coordination Challenges	
Speed and reaction time	*Tetris*
Accuracy or precision (steering, shooting)	*OutRun, Rainbow Six*
Timing and rhythm	*Just Dance*
Learning combination moves	*Street Fighter II*
Formal Logic Challenges	
Deduction and decoding	*Minesweeper, Mastermind*
Pattern Recognition Challenges	
Static patterns	*Brain Age, Bejeweled*
Patterns of movement and change	*Sonic the Hedgehog,* behavior patterns of enemies
Time Pressure	
Beating the clock	*Frogger, Diner Dash*
Achieving something before someone else	*Need for Speed*

Memory and Knowledge Challenges	
Trivia	*You Don’t Know Jack*
Recollection of objects or patterns	*Brain Age*
Exploration Challenges	
Identifying spatial relationships	*Descent,* navigating in three dimensions
Finding keys (unlocking any space)	*Ultima*
Finding hidden passages	*Doom*
Mazes and illogical spaces	*Zork*
Conflict	
Strategy, tactics, and logistics	*Warcraft,* commanding armies
Survival	*Pac-Man,* avoiding being caught
Reduction of enemy forces	*Space Invaders,* killing aliens
Defending vulnerable items or units	*Ico,* looking after a little girl who can’t fight
Stealth	*Assassin’s Creed,* avoiding being seen

Economic Challenges	
Accumulating resources or points (growth)	*Civilization*
Establishing efficient production systems	*The Settlers*
Achieving balance or stability in a system	*SimEarth*
Caring for living things	*The Sims*
Conceptual Reasoning Challenges	
Sifting clues from red herrings	*Law and Order,* solving crimes
Detecting hidden meanings	*Planescape: Torment,* understanding characters’ motivations from vague hints
Understanding social relationships	*Façde,* reconciling a quarreling couple
Lateral thinking	*The Incredible Machine,* building a machine from limited parts
Creation/Construction Challenges	
Aesthetic success (beauty or elegance)	*The Sims,* assembling a photo album
Construction with a functional goal	*Minecraft*

FINAL ASSIGNMENT

UNIVERSIDADE da MADEIRA

In groups of up to 5, create a fun and engaging digital game (only one level required) that challenges the unrealistic beauty standards often portrayed on social media, such as the idea that being skinny or looking a certain way is the only way to be beautiful.

Your game should:

- Be enjoyable and interactive for players.
- Show how social media can influence how young people view themselves.
- Promote self-acceptance, confidence, and a more inclusive idea of beauty.
- Encourage players to reflect on what real self-esteem means beyond filters and follower counts.
- You can use humour, storytelling, characters, or challenges… be creative! The goal is to entertain while making players think and feel empowered.

- You should also conduct a pilot study and reflect on the feedback for future refinement and iterations.

USER TESTING

UNIVERSIDADE da MADEIRA

Measuring and evaluating the usability of interactive systems and interfaces is an ongoing challenge for teams that are concerned with improving the user experience (Finstad, 2010).

Numerous usability evaluation methods can be used to estimate the usability problems at the earlier stage of software design in order to overcome the severity level and withdraw the rework of design (Djamasbi & Wilson, 2017; Finstad, 2010; Sauro, 2015).

https://www.youtube.com/watch?v=C-Ym5zlU9Uk

USER TESTING

UNIVERSIDADE da MADEIRA

Game Engagement Questionnaire (GEQ),

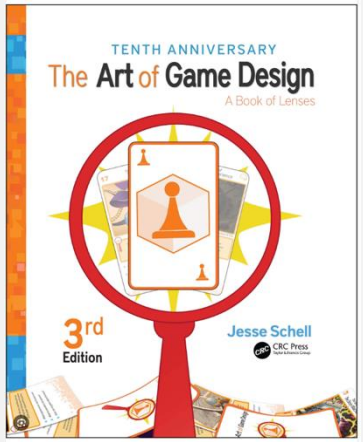
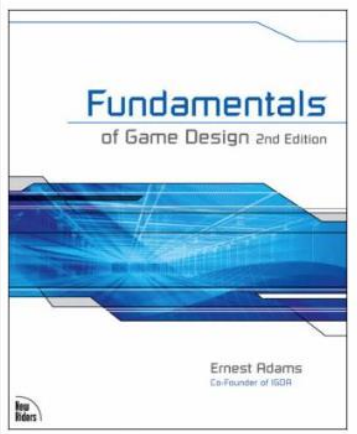
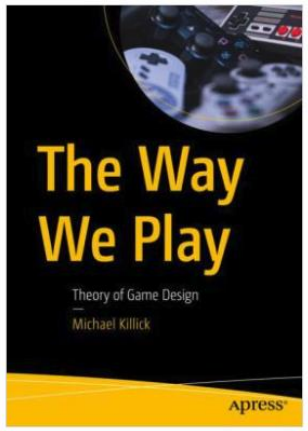
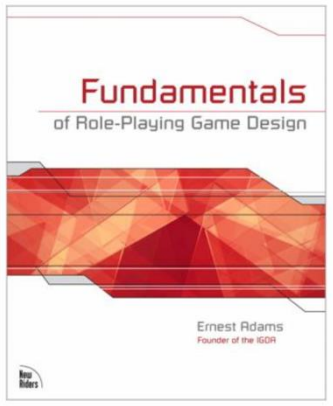
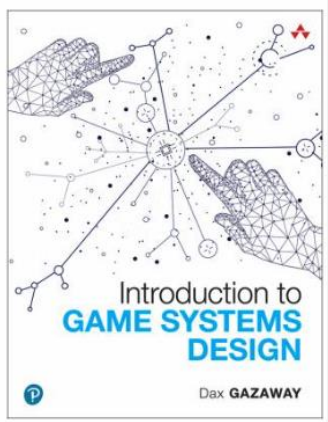
System Usability Scale (SUS),

Players Experience Inventory (PEI),

Player Experience of Need Satisfaction (PENS),

Intrinsic Motivation Inventory (IMI),

Rosenberg Self-Esteem Scale (impact)

REFERENCES

UNIVERSIDADE da MADEIRA